APAN DINDA

PhD Student
at Depertment of Physics
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Summary

I am a PhD student in the Department of Physics at the Indian Institute of Technology Hyderabad (IIT H) under the supervision of Dr. Archak Purakayastha. My keen interest extends to theoretical aspects and the practical implementation of these concepts in experimental settings. My current research field is "Driven Dissipative Quantum Systems".

Education

Degree	Institute	Board / University	CGPA/Percentage	Year
PhD	Indian Institute of Technology, Hyderabad	IIT Hyderabad	9.5	2024 - ongoing
Masters(M.Sc. Physics)	Indian Institute of Technology, Hyderabad	IIT Hyderabad	9.23	2022-2024
Bachlors(B.Sc.(H) Physics)	Panskura Banamali College (Autonomous)	Vidyasagar University	8.97	2019-2022
Senior Secondary(major:PCMB)	Shyamsundarpur Patna High School	WBCHSE	87.8%	2018-2019
Matriculation	Shyamsundarpur Patna High School	WBBSE	89.2%	2017

Area of Interest

- Open Quantum System
- Quantum Thermodynamics
- Non Equilibrium Physics
- Quantum Optics
- Theoretical Condensed Matter Physics and Topological aspects

Achievements

- *Mar, 2025 - July, 2025:* **JST Fellow** for an exchange program called 'Young Indian Researchers Invitation Program', funded by Japan Science and Technology (JST).
- *Dec 2024:* **CSIR-NET PhD fellowship:** Secured **all India rank 80 (out of 26598 candidates)** in Council of Scientific and Industrial Research National Eligibility Test (CSIR NET) examination.
- *Feb,2024-Jun,2024:* **Postgraduate Chanakya Fellowship from I-Hub Quantum Technology Foundation:** Fellowship for carrying out M.Sc project on "Engineering Gain Via Quantum Measurements", with Dr. Archak Purkayastha.
- *21 March, 2024:* **Academic Excellence Award** at Indian Institute of Technology Hyderabad for securing highest CGPA.
- *2021-2022:* **Secured all India rank 355 (out of 12740 candidates) in Joint Admission test for Masters (JAM) for Indian Institute of Technology (IIT).**
- *2017-2019:* Earned **Jagadis Bose National Science Talent Search (JBNSTS) Scholarship,** prestigious national level scholarship examination for inspiring students towards research in basic science.
- *2014-2016:* Received **State-level merit cum means scholarship,** State-level scholarship examination, financial support to the extraordinary students in studies.

Projects

- **Lack of unique steady state in collisional models of cavity-qubit interactions** *Ongoing*
  Supervisor: Dr. Archak Purkayastha, Assistant Professor, IITH
  – Study of qubit-cavity interaction of Jaynes-Cummings type based on Quantum Collisional models
  – Mapping to inhomogeneous random walk and study of deviation from continuous Lindblad dynamics

- **Engineering Gain Via Quantum Measurements (Master's Project)** *July 2023 - May 2024*
  Supervisor: Dr. Archak Purkayastha, Assistant Professor, IITH
  – Study of a gain medium of single qubit connected with bosonic mode using directly by Jaynes Cummings Model and via Lindblad form based on Quantum Collision Model.
  – From various initial conditions of both qubit and bosonic mode gain simulation to investigate the non-linear behaviors.

– Analysis of the states using Second Correction function and Wigner Quasi-probably Distribution.

- **Dynamics and thermodynamics of driven resonant level (Group Project)** *May 2023 - 2024*
  Supervisor: Dr. Archak Purkayastha, Assistant Professor, IITH
  – Study of a resonant level coupled to multiple baths at different temperatures and chemical potentials.
  – Energy current, transferred heat, and efficiency calculation in wide band limit.

Technical Skills

- **Programming Languages**: ***Python*** (numpy, matplotlib, scipy, sympy)
- **Useful Application**: ***Mathematica, Matlab, Latex***

Key courses taken

- **Open Quantum System**: Quantum Statistics, Quantum Thermodynamics, Noisy system, Quantum Super Operators and dynamics, Quantum Regression, Quantum measurements, Quantum Collision Model.
- **Gravitation and Cosmology**: Newtonian Gravity, Point group Topology, Differential Geometry, Fierz-Pauli Equation, Geodesics, and its deviation equation, Schwarzschild metric and solution, Einstein Field Equations and its verification, Introduction to cosmology.
- **Non-Equilibrium Many-Body Physics**: Non-interacting many-body quantum system with second quantization, Green's function approach and various correlation functions, linear response theory, Anderson impurity model, and the Quantum Ising model, Floquet theory.
- **Computational Solid State Physics**: Use of Quantum Espresso for Condensed matter system (DFT).

Languages

- English
- Bengali
- Hindi